

MATH 1342 – Elementary Statistics – Week 12: Inference for Two Proportions and for a Mean

Nov 10 – 12 | 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) • 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#	
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CO8) 2. Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CO9) 3. Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CO10) 4. Explain why inference for a mean uses the t distribution with $df = n - 1$, and check the conditions for t procedures. (CO4) 5. Construct and interpret a t confidence interval for a population mean, and carry out a one-sample t test. (CO4, CO9, CO10)	e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT)	MO1, MO2, MO3	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	MO1, MO2, MO3	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM)	MO1, MO2, MO3	
		Interactive Lecture: 6.3 Inference for a Difference in Proportions	MO1, MO2, MO3					
		Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	MO1, MO2, MO3	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	MO4, MO5	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)	MO4, MO5	
		e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT)	MO4, MO5					
		Interactive Lecture: 6.2 Inference for a Mean	MO4, MO5			Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 5.1, 6.1–6.3	All	
		Slides: 6.2 Inference for a Mean	MO4, MO5					
		StatKey – Theoretical Distributions (t)	MO4, MO5					